

Anomaly Detection in Traffic Data

First Author Name^{a*}, Second Author Name^{bc}, etc.^{ac}

^aDepartment, University, City, Country

^bDepartment, University, City, Country

^cDepartment, University, City, Country

*Corresponding author: first name, initials, surname; email.address

Abstract

1 Introduction

The continuous distribution and utilization of traffic data by various organizations underscore the necessity for robust analysis to identify and address potential anomalies, ensuring the integrity of the data disseminated. Analyzing such data and detecting anomalies is crucial for maintaining the accuracy and reliability of traffic information systems. Anomaly is a data point or pattern that significantly deviates from the expected behavior or norm in a dataset. Anomaly detection techniques, particularly those leveraging machine learning models, offer promising solutions for this task, for example when applied to time-series traffic data. However, the challenge amplifies when implementing these techniques across a network of road detectors distributed spatially over a region. These detectors collect vital traffic metrics such as speed, flow, and occupancy. Effectively using the spatial relationships and correlations among these detectors is essential for accurate anomaly detection.

In this paper, we propose a set of methods, which will be used collectively to detect anomalies in traffic data. Our approach utilizes data collected by road detectors, i.e. flow, speed, and occupancy values recorded at two-minute intervals. By analyzing this data, we aim to identify potential anomalies in the detectors on a particular day. Our methods leverage the spatial distribution and interrelationships among the detectors, as well as machine learning models, to provide a robust and accurate system for anomaly detection.

The problem statement is to utilize the road traffic data captured by a network of known detectors, and predict whether we can find an anomaly in any of the detector.

Contributions and paper organization:

- **Triplet Formation and Anomaly Detection (Section 2.1):** We propose a novel approach for forming triplets based on nearby detectors. By calculating the correlation coefficients among these triplets, we effectively identify anomalies within the data. This method allows us to curate a dataset comprising only high-quality data, which is crucial for accurate profile training.
- **Profile Model Training (Section 2.2):** Utilizing the curated dataset from the previous step, we train a profile model using a classification algorithm. This model processes daily volume data, sampled at a 2-minute frequency, and outputs labels corresponding to the respective detectors. The aim is to establish robust profiles that reflect normal operational patterns.
- **Supervised Binary Classification for Anomaly Detection (Section 2.3):** In this section, we develop and train a supervised binary classification model. The model is trained on labeled data, where labels are determined based on correlation values. The input features are derived from the triplet data, enabling the model to classify anomalies effectively.

- **Results (Section 4) and Conclusion (Section 5):** We present a comprehensive analysis of the results obtained from the proposed methods, highlighting the effectiveness and accuracy of our approach. Finally, we conclude the paper by summarizing the key findings and discussing potential directions for future research.

1.1 Related Work

This subsection will contain the literature review.

2 Methodology

2.1 Dataset Description

This study focuses on the analysis of traffic data in Hong Kong, utilizing datasets obtained from the official Hong Kong government website. A comprehensive network of cameras (detectors) has been strategically deployed across major road routes in Hong Kong. These detectors capture images and subsequently process them to generate lane-wise data on speed, flow, and occupancy at each monitoring station.

The dataset spans a **six-month period**, from October 2023 to March 2024, with data sampled at a high frequency of every 2 minutes. This extensive dataset comprises the flow, speed, and occupancy measurements from 768 detectors, recorded across multiple lanes for the entire duration. Given the raw nature of this data, it contains numerous null values and anomalies, necessitating preprocessing and anomaly detection techniques.

In addition to the traffic data, the government website provides detailed metadata on the exact locations of **786 detectors**. This metadata includes the detector ID or name, latitude and longitude coordinates, the name of the road where the detector is situated, and the direction of the traffic flow that each detector captures (bound direction). The direction of the traffic flow is critical as different detectors are positioned to monitor different directional flows on the same road.

We leverage this spatial information to enhance our network-based analysis. By integrating the geographic coordinates and the directional flow data, we can perform a more detailed and context-aware examination of the traffic patterns, which is essential for understanding the underlying dynamics of traffic flow and for detecting anomalies within the network.

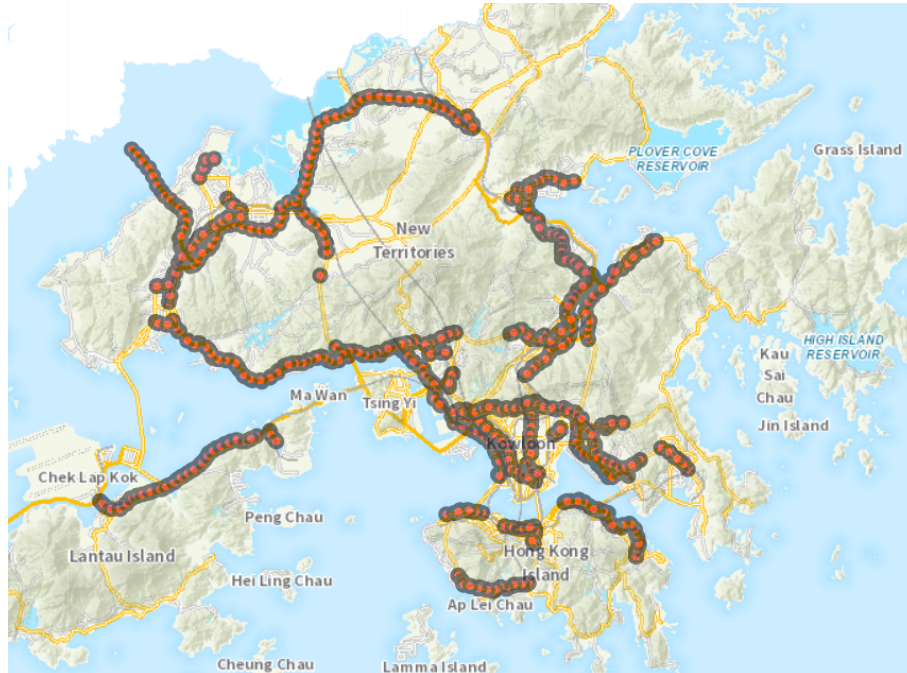


Figure 1: Location of Traffic Detectors in Hong Kong

2.2 Correlation of Nearby Detectors

Here we explain why correlation of nearby detectors needs to be high, hence using this metric to pick out anomalies.

Then using the naming of detectors and the location dataset, explain in detail how we form triplets of detectors which are nearby (also mention how we handle null values). So in a triplet the vehicles are flowing in one direction.

First using the correlation in the triplet detectors to identify 3 types of detectors: ≥ 0.9 , $\geq 0.6 \& \leq 0.9$, ≤ 0.6 and explaining their meaning and how we chose the cutoff value.

2.3 Profile Model of Detectors

Here using the correlation values, now we calculate the correlation among the triplets for each day, to obtain a dataset where columns are detector names, rows are days and the data is the correlation of that detector on that day with nearby ones. If the value is ≥ 0.9 we consider that data point into the training data for profile model.

Explain the architecture and other details of ML model which is a classification model. Inputs are the 2 min-freq flow values of a particular day of a particular detector, and the output is the label of the predicted detector.

Then we explain how we set thresholds on rank of the predicted detector for the accuracy. then we check the relation between correlation values and the rank of correct detector. we first compare on common time axis the value of accuracy and the correlation, where we find similar trend in both. Then next we compare the rank vs accuracy graph and how we set thresholds.

2.4 XYZ method

3 Results

3.1 Correlation of Nearby Detectors

Here we show sample time series graphs of the 3 sections of the correlation values, and justify each.

3.2 Profile model

Here we show all the training plots and metric values first. Then we show plots comparing the results of profile model with the correlation coefficient and show samples to justify it.

4 Conclusion

5 References